

Guessing or Understanding? Probing the Robustness of Large Language Models in Mental Health Sensing

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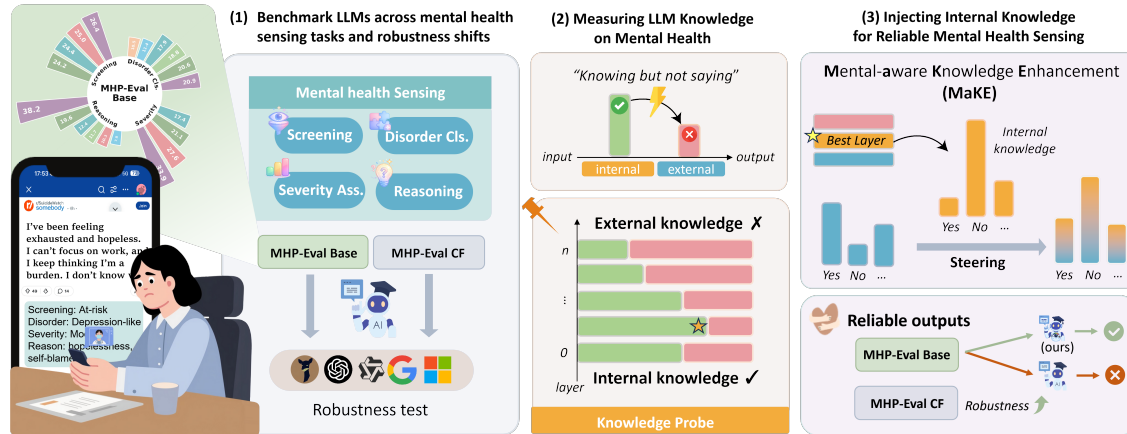


Fig. 1. This figure summarizes our pipeline for evaluating and improving large language models for mental health sensing from online text. We benchmark model performance and counterfactual robustness (Sec. 4), probe internal versus external mental health knowledge (Sec. 5), and then steer model outputs using internal knowledge to produce more reliable predictions (Sec. 6).

We study the robustness of large language models (LLMs) for mental health sensing and expose a critical *knowing-but-not-saying* mismatch between what models encode and what they ultimately express. We introduce MHP-Eval, an evaluation framework covering four common sensing needs—screening, disorder classification, severity assessment, and reasoning—and systematically stress-test robustness with meaning-preserving counterfactual perturbations that change surface form. Our results show that apparent competence can mask fragile behavior: performance varies substantially across tasks, and small counterfactual edits can shift clinically relevant judgments. To move beyond end-task accuracy, we separate external knowledge (final outputs) from internal knowledge (model-internal evidence from hidden representations), finding that internal evidence is often more stable and can anticipate failures. Building on this insight, we propose MaKE, a training-free inference-time reliability strategy that leverages internal evidence to retrieve task-relevant support and correct brittle decisions. Under counterfactual perturbations, MaKE improves robustness by +0.3121 on screening, +0.1619 on disorder classification, and +0.257 on severity, enhancing reliability without additional model training.

CCS Concepts: • **Human-centered computing** → **Ubiquitous and mobile computing**.

Additional Key Words and Phrases: Mental LLM, Mental Health Detection, LLM-training-free Inference

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Manuscript submitted to ACM

Manuscript submitted to ACM

ACM Reference Format:

Anonymous Author(s). 2018. Guessing or Understanding? Probing the Robustness of Large Language Models in Mental Health Sensing. In *Proceedings of Make sure to enter the correct conference title from your rights confirmation email (Conference acronym 'XX)*. ACM, New York, NY, USA, 29 pages. <https://doi.org/XXXXXXX.XXXXXXX>

1 Introduction

Large language models are beginning to shape how people talk about mental health in everyday life. Users increasingly describe stress, low mood, panic, or burnout in a chat interface and ask for feedback, explanation, or next steps [35, 56]. This shift matters because mental health conversations often occur when people feel uncertain, overwhelmed, or isolated, and a fast, fluent reply can be unusually persuasive [51]. At the same time, mental health is a high-stakes domain: a response that is confident but wrong may mislead a user about their condition, discourage appropriate help-seeking, or reinforce harmful beliefs [16, 35]. As access barriers continue to drop, model outputs are more likely to influence real decisions, making careful evaluation a practical requirement rather than a purely academic concern [16, 35].

In parallel, online text remains a valuable signal for mental health sensing. Social media posts and forum discussions are low-burden and in-the-wild, capturing how people narrate their experiences over time with fewer constraints than surveys [8, 19?] or clinical interviews [17, 23]. As a result, mental health sensing from online text is increasingly relevant not only to researchers but also to deployed systems and everyday use, including cases where users paste personal text into a chatbot and treat the response as an informal assessment [70]. Across health informatics, clinical AI, and HCI, reviews have therefore raised repeated concerns about safety, ethics, and reliability when LLM outputs are treated as mental health guidance [10, 31, 62].

These trends create a central challenge for HCI and AI. Mental health sensing requires judgments that are dependable under natural variation in language: if a user describes the same situation in slightly different words, or if a post is edited without changing its meaning, the inference should not shift in ways that change the implied risk or recommended action. Yet fluent outputs alone do not guarantee grounded understanding. An LLM may produce a plausible label or explanation without using mental health concepts in a stable way, or it may rely on superficial patterns that correlate with labels in data rather than reasoning about the underlying construct. This motivates two questions that sit at the core of our work:

- **RQ1:** How do Large Language Models perform across the spectrum of mental health sensing tasks, including Screening, Disorder Classification, Severity Assessment, and Clinical Reasoning, and how robust are their capabilities against counterfactual perturbations?
- **RQ2:** Do Large Language Models genuinely possess mental health knowledge when addressing mental health detection tasks?

These questions also motivate a third. If LLM behavior is unstable or incorrect, it is not sufficient to report performance gaps; we need evidence that helps explain when and why failures occur, and we need practical ways to improve reliability in settings where repeated retraining may be costly or infeasible. This leads to:

- **RQ3:** Can we enhance the reliability and robustness of Large Language Models in mental health sensing based on the insights into their internal mechanisms?

To address these questions, we introduce **Mental Health Probing Evaluation (MHP-Eval)**, an evaluation framework that treats mental health sensing as more than a single label prediction. MHP-Eval covers four task types that reflect common sensing needs (Screening, Disorder Classification, Severity Assessment, and Reasoning), and stress-tests

robustness using *counterfactual perturbations* that preserve the underlying meaning while altering surface form. Beyond end-task accuracy, we explicitly separate what a model encodes internally from what it expresses externally: we measure *external knowledge* from the model’s final responses, and *internal knowledge* from model-internal evidence (e.g., layer-wise representations) that reveals whether the correct decision is supported in the hidden states even when the output is incorrect. Building on this internal–external gap, we propose **Mental-aware Knowledge Enhancement (MaKE)**, a *training-free* reliability strategy that leverages internal evidence to retrieve task-relevant support and correct brittle decisions at inference time.

Our results show that apparent competence can mask fragile behavior. Performance varies substantially across task types, and counterfactual edits can shift model judgments in ways that matter for real-world sensing. Importantly, internal evidence often remains stable even when the final response flips, indicating a *knowing-but-not-saying* failure mode that standard evaluation overlooks. **MaKE** turns this insight into practical gains: under counterfactual perturbations, it improves robustness by **+0.3121** on Screening, **+0.1619** on Disorder Classification, and **+0.257** on Severity, demonstrating that reliability can be enhanced without additional model training.

2 Related Work

2.1 Passive Sensing of Mental Health via Online Text

Online text traces from social media and online forums provide a low-burden, in-the-wild signal for passive sensing of mental health [8, 17, 19, 23], offering a complementary lens to traditional self-reports [33] and clinical assessments [2]. Over the past decade, substantial work has leveraged user-generated text to identify or monitor risks related to a broad range of mental health conditions, including depression, anxiety, PTSD, suicidal ideation/self-harm, stress, and eating disorder [6, 11, 14, 40, 60, 68, 69]. The real-time and archival nature of online platforms enables longitudinal observation of individuals’ expressions and experiences, supporting risk monitoring and, in some settings, timely identification of heightened-risk signals in naturalistic contexts [14, 40, 69]. Online text traces have become an increasingly important and scalable resource for mental health research; as online LLMs become a common venue where people discuss stress, anxiety, and related concerns in text form [4, 18], the need for rigorous, clinically grounded inference from such naturalistic expressions is further amplified.

To facilitate reproducible evaluation, the community has developed benchmarks and shared tasks for mental health inference from online text, such as CLPsych [14] for depression and PTSD and eRisk [40] for early risk prediction, alongside widely-used multi-condition corpora such as SMHD [11] and depression-focused resources in the spirit of RSDD-style datasets [68], as well as domain-focused datasets such as Dreddit for stress [60]. Prior to the rise of LLMs, mental health sensing primarily relied on supervised classification. Early methods utilized linguistic features to quantify associations with mental conditions [13, 17, 68], which later evolved into deep learning approaches that learned representations directly from text [6, 28, 57, 69]. However, prior research warns that high benchmark accuracy often masks validity issues—such as reliance on data biases rather than true psychological inference [8, 19, 25, 50]. As LLM is increasingly used for real-world support [5, 54], addressing these limitations is crucial. Therefore, our work moves beyond black-box accuracy to investigate whether LLMs truly grasp mental health constructs or merely rely on superficial patterns.

2.2 Mental Health LLMs

Large language models are increasingly used in mental health related settings, from everyday self disclosure in chat to structured support and triage. Recent HCI evidence shows that people already engage with LLM chatbots for mental

health needs and share sensitive experiences through natural conversation [56]. At the same time, reviews across HCI, health informatics, and clinical AI have documented rapid growth of LLM based mental health applications, while repeatedly warning about safety, ethics, and reliability risks when model outputs are taken as mental health guidance [10, 31, 62, 70]. This combination of rising real world use and high stakes consequences makes it important to evaluate what these models can and cannot do in mental health contexts, especially whether their outputs reflect stable understanding of psychological concepts rather than convenient language matching.

In parallel with this adoption, research has quickly split into two threads. One thread adapts LLMs for mental health sensing and detection from online text through instruction following. For example, *Mental-LLM* [65] studies multi-task instruction tuning for social media mental health prediction and reports gains over zero-shot prompting, and *MentalLaMA* further trains an instruction-following model that performs multi-task social media mental health analysis with explanation-style outputs [67]. Related efforts also explore lighter or domain-specialized mental health LLMs for classification tasks, such as *MentalQLM* [53]. A second thread targets support: models are fine tuned on counseling transcripts to generate more appropriate guidance [34, 38], specialized for therapy styles such as CBT [44], or further improved with reinforcement learning and structured emotional reasoning for supportive dialogue [52]. Across both threads, most evaluations emphasize average task scores or perceived response quality, leaving open a core question for mental health sensing: *whether an LLM is making a construct level inference grounded in mental health knowledge, or whether it is mainly producing plausible guesses driven by superficial correlations in the data*. Our work is positioned to answer this question by testing robustness to counterfactual changes, probing whether knowledge is stable across repeated measurements, and connecting these behaviors to internal model mechanisms to inform reliability improvements.

2.3 Knowledge of LLMs

The knowledge of language models has been studied extensively, with many works evaluating whether a model “knows” information by testing recall in cloze-style completion or question answering, either directly in a zero-shot setting or after fine-tuning [12, 15, 48, 55, 63, 74]. LLMs can encode *hidden knowledge* in their internal activations, which can be revealed by probing or extraction methods even when the model’s final outputs are unreliable [3, 7, 42, 46]. More recently, researchers have proposed clearer operational definitions of knowledge that go beyond treating a single answer as evidence. For example, Gekhman *et al.* [21] formalizes knowledge as the model’s ability to rank correct answers above incorrect ones for a given question, and uses this definition to distinguish what the model expresses externally from what it may encode internally. Inspired by this work, we examine whether LLMs exhibit stable mental health knowledge when performing mental health sensing from online text.

3 MHP-Eval: A Holistic Benchmark for Online Text Mental Health Detection

To systematically investigate our research questions—specifically, to distinguish whether LLMs are merely guessing based on surface patterns or truly understanding clinical concepts—we require a rigorous and holistic evaluation framework. While prior studies have explored various mental health detection tasks, they often rely on fragmented evaluations that fail to cover the full spectrum of sensing needs or neglect the critical dimension of model robustness. To bridge this gap, we introduce the **Mental Health Probing Evaluation (MHP-Eval)** benchmark.

As illustrated in Figure 2, MHP-Eval is designed to stress-test LLMs through a unified pipeline consisting of two complementary components:

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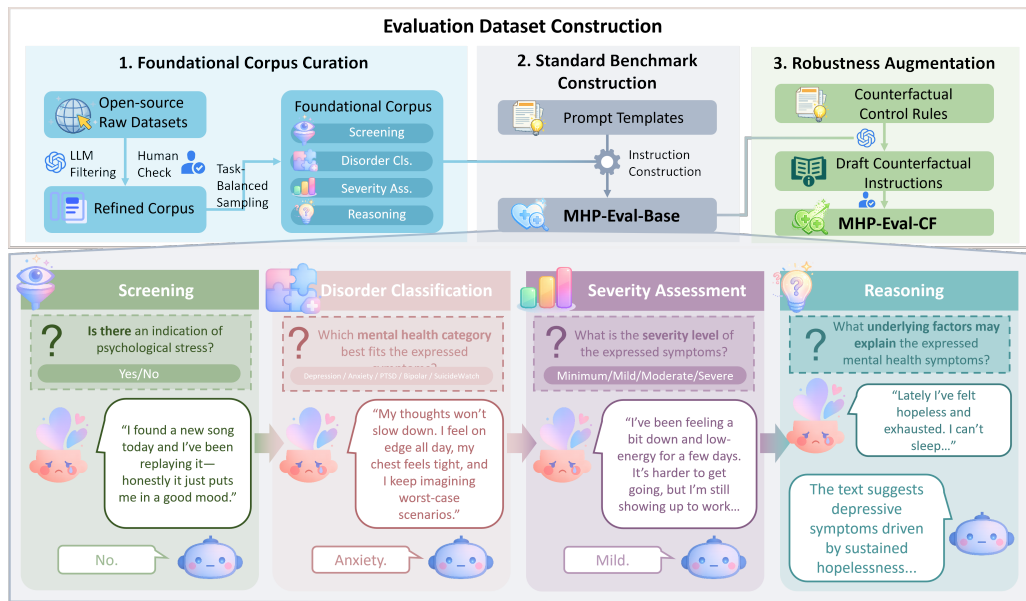


Fig. 2. Construction pipeline and task examples of **MHP-Eval**. The upper panel details the three-stage dataset creation process. The lower panel displays representative samples for the four sensing tasks.

- **MHP-Eval-Base:** A standardized benchmark that integrates four distinct tasks—Screening, Disorder Classification, Severity Assessment, and Reasoning—to evaluate general capabilities across different stages of mental health analysis.
- **MHP-Eval-CF (Counterfactual):** A robustness-focused extension containing counterfactually augmented samples. This component is designed to verify the stability of model capabilities, ensuring that predictions rely on genuine clinical understanding rather than superficial correlations.

3.1 Construction of MHP-Eval-Base

The construction of MHP-Eval-Base focuses on transforming raw, noisy social media data into a high-quality, instruction-following benchmark. As illustrated in the upper panel of Figure 2, this process consists of two key phases: foundational corpus curation and instruction construction.

3.1.1 Foundational Corpus Curation. To ensure our benchmark covers a diverse range of mental health conditions and expression styles, we aggregated data from eight widely recognized open-source datasets. These sources were selected to provide broad coverage across our four target tasks:

- **Broad Mental Health & Disorder Classification:** We incorporated **SWMH** [29], **Reddit Mental Health Dataset** [41], **SDCNL** [24], and **IMHI** [67]. These datasets provide rich, labeled posts covering conditions such as depression, anxiety, PTSD, and bipolar disorder. Notably, given that **IMHI** contains extensive expert-annotated explanations, we specifically utilized it as the primary source for the ground truth references in our Reasoning task.
- **Stress & Severity Assessment:** We utilized **DepSeverity** [45], **Dreaddit** [60], and **SAD** [43], which offer fine-grained annotations regarding the intensity of symptoms and stress levels.

- **Suicide Risk:** We included the **Reddit C-SSRS Suicide Dataset** [20] to ensure our benchmark captures high-stakes critical risk signals.

Filtering and Quality Control. Raw social media text is inherently noisy. Direct usage of such data often leads to invalid evaluations. To address this, we implemented a rigorous two-step filtering pipeline involving both LLM-based automated cleaning and human verification:

- (1) **Noise Removal via LLM:** We utilized GPT-5 to filter out posts containing severe linguistic noise that hinders clinical interpretation. Based on our preliminary analysis, we targeted four primary noise patterns: illegible gibberish or scrambled spelling (e.g., “sni ful ik struggling” instead of “sniffing and struggling”), structural incoherence due to missing punctuation (e.g., run-on sequences like “worry daughter taken away crazy ex wife”), semantic unintelligibility arising from chaotic grammar (e.g., “lost self know anymore”), and content dominated by meaningless repetition (e.g., “think think goal ambition”). This semantic-level filtering ensured that the retained samples possessed sufficient linguistic clarity for valid psychological inference.
- (2) **Human Verification:** Five trained annotators reviewed the filtered subset to ensure the remaining texts retained clear semantic meaning and matched their original labels.

Through this process, we removed approximately 15% of the raw data that was deemed unsuitable for evaluation. Finally, to ensure fair evaluation, we performed task-balanced sampling, selecting exactly 2,500 high-quality samples for each of the four tasks, resulting in a total of 10,000 samples for MHP-Eval-Base (see Figure 4 for the detailed distribution).

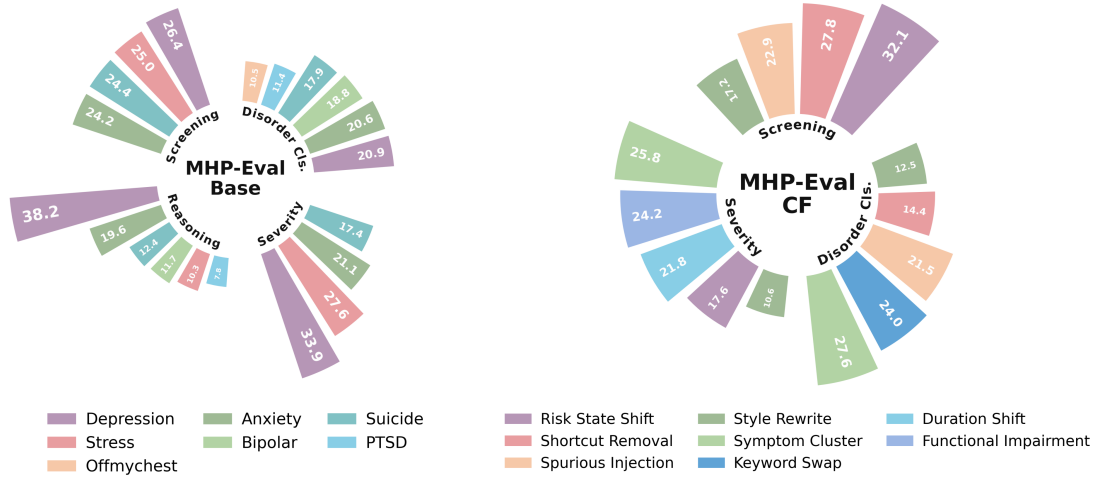


Fig. 3. Dataset distribution. **(Left)** Composition of MHP-Eval-Base across tasks and mental health categories. **(Right)** Breakdown of MHP-Eval-CF across tasks and counterfactual perturbation types

3.1.2 Standard Benchmark Construction via Prompt Templates. To evaluate LLMs in an instruction-following setting, we transformed the raw (Text, Label) pairs into structured (Instruction, Input, Output) triples. A key design consideration was robustness to prompt phrasing (such as Mental-LLM [65]). A model should not succeed merely because it memorized a specific question format. Therefore, for each task, we designed 5 distinct prompt templates with varying phrasing

Table 1. Taxonomy of the 8 counterfactual operations used in MHP-Eval-CF. We group them into Label-Preserving (LP) and Label-Flipping/Directional (LF) types. Each row shows a definition (gray) and a minimal edit example with deletions (red) and insertions (green).

Operation	Definitions and Examples
Label-Preserving (LP): Robustness to Surface Variation	
Style Rewrite	Rewrite expression (lexico-syntactic paraphrase) preserving facts/label. I'm so sad today. → A wave of melancholy has washed over me today.
Shortcut Removal	Remove explicit shortcut cues (diagnosis/source) preserving evidence. My depression is bad. → My constant low mood and hopelessness is bad.
Keyword Swap	Replace high-trigger keywords with symptom-grounded descriptions. I feel anxious. → My heart races and I can't stop worrying.
Spurious Injection	Insert non-causal cue, keeping main content unchanged. I can't focus. → I can't focus. (Posted in r/depression.)
Label-Flipping (LF): Sensitivity to Clinical Signals	
Duration Shift	Change frequency/duration expression (severity driver). Felt low for a few days. → Felt low nearly every day for two months.
Functional Impairment	Add/remove functional impairment evidence (work/school/social impact). I'm stressed. → I'm stressed, and I missed work twice this week.
Risk State Shift	Change risk-state fact (e.g., suicidal ideation) without extra factors. I'm overwhelmed, not thinking of suicide. → I'm overwhelmed, thinking of suicide.
Symptom Cluster	Modify symptom set/quantity while keeping duration constant. I can't sleep. → I can't sleep, and I can't focus.

but identical semantic intent. During benchmark construction, one template was randomly selected for each sample. Table ?? illustrates examples of these variations.

3.2 Construction of MHP-Eval-CF

While MHP-Eval-Base measures general performance, prior NLP research suggests that high accuracy on static test sets is an incomplete proxy for robust semantic understanding, since LLMs can exploit superficial patterns and remain brittle under counterfactual or minimally-edited inputs that preserve the underlying meaning [58, 76]. It remains an open question whether LLMs in mental health sensing similarly depend on superficial triggers—such as self-diagnosis keywords or platform-specific metadata—instead of capturing clinical constructs. To systematically probe these decision boundaries, we construct **MHP-Eval-CF** (illustrated in Figure 4), a counterfactual evaluation suite designed to disentangle superficial pattern matching from reliable clinical reasoning.

3.2.1 Motivation and Scope. The core goal of MHP-Eval-CF is to assess *local robustness* by measuring decision stability under perturbations. We apply this extension to the three discriminative tasks: **Screening**, **Disorder Classification**, and **Severity Assessment**.

Exclusion of Reasoning Task. We deliberately exclude the Reasoning task from this counterfactual expansion. Evaluating the correctness of counterfactually generated *explanations* remains an open challenge; current metrics (e.g., BLEU, BERTScore) are insufficiently sensitive to subtle logic shifts required by counterfactuals, and defining a "ground truth" explanation for a perturbed input introduces significant subjectivity.

3.2.2 *Design Principles.* To ensure the validity of our counterfactuals, we adhere to three design principles inspired by prior work in counterfactually augmented data (CAD) [9, 64, 71]:

- (1) **(P1) Minimal Intervention:** Each counterfactual modifies only a single targeted factor (e.g., duration, symptom intensity) while preserving the remaining context.
- (2) **(P2) Consistent Label Behavior:** The perturbation must have a deterministic effect on the label—either preserving it (Invariance) or shifting it logically (Directional Expectation).
- (3) **(P3) Semantic Coherence:** The generated text must remain fluent and clinically plausible, avoiding robotic phrasing or logical contradictions.

3.2.3 *Counterfactual Taxonomy.* Based on these principles, we implemented eight distinct counterfactual operations, categorized into **Label-Preserving (LP)** and **Label-Flipping (LF)** groups. The specific definitions and examples are detailed in Table 1. We adopted a rigorous human-in-the-loop pipeline to ensure the high quality of the counterfactual samples (refer to the right panel of Figure 2). This process consists of two stages:

Stage 1: Rule-Guided Automated Generation. First, we leveraged GPT-5 to produce initial counterfactual candidates based on the taxonomy defined in Table 1. For each source sample (x, y) in MHP-Eval-Base, we constructed specific prompts corresponding to the target operation. For instance, a prompt might instruct the model to rewrite the text to change the symptom duration from 'days' to 'years' while keeping other details unchanged. This step allowed us to rapidly scale the generation of potential counterfactual pairs (x, x') that theoretically satisfy the label-preserving or label-flipping criteria.

Stage 2: Human Refinement and Verification. Since LLM-generated edits can occasionally be excessive, grammatically unnatural, or fail to adhere to the minimal edit principle, human verification is essential. We recruited 10 annotators with background knowledge in psychology to review the candidates. To facilitate this, we developed a custom web-based annotation interface, as visualized in Appendix Figure 10. During this phase, annotators were presented with the original and perturbed text side-by-side. They were instructed to rigorously check each pair against three criteria. Specifically, they first verified whether the perturbation accurately reflects the intended operation, then ensured the edit is minimal with no unnecessary changes to non-causal spans, and finally confirmed that the text remains semantically coherent. Annotators could either *accept* valid pairs, *manually edit* imperfect ones to fix logical or grammatical flaws, or *reject* samples that were unsalvageable. The final MHP-Eval-CF dataset consists only of samples that passed this strict verification process.

4 RQ1: Benchmarking LLMs on Mental Health

4.1 Evaluation Setup and Metrics

Baseline Models. To evaluate the performance of LLMs in mental health tasks, we select a diverse set of models to compare general capabilities with domain-specific expertise. This benchmark covers a wide range of baselines, including open-source generative models, proprietary state-of-the-art models, and traditional deep learning classifiers. Our primary focus lies on *Open-source General LLMs*, covering the technological evolution from 2023 to late 2025. We selected 10 representative models ranging from 1B to 8B parameters, including the **Qwen** series (**Qwen2.5-1.5B/3B/7B** [49], **Qwen3-4B/8B** [66]), alongside widely-adopted lightweight models such as **Gemma-3-1B** [59], **Llama-3.1-8B** [22], **Mistral-7B-v0.3** [32], **Phi-3-mini-4k(4B)** [1], and **Vicuna-7B-v1.5** [75]. To benchmark these against the industry upper bound, we incorporate *Closed-source LLMs*, specifically **GPT-4o** and **GPT-5**. Furthermore, to investigate the

efficacy of domain adaptation, we evaluate specialized *Open-source Mental-Health LLMs*, including **MentallLM** [65] and **MentalLaMA** [67].

For the discriminative tasks (i.e., Screening, Disorder Classification, and Severity Assessment), we extend our comparison to traditional *Mental Health Deep Learning Models*. This category includes encoder-only architectures such as **Mental-BERT** and **Mental-RoBERTa** [30], as well as biomedical foundations like **Bio-ClinicalBERT** [36] and **ClinicalBERT** [27].

Evaluation Metrics. We employ distinct metrics tailored to the nature of each task:

- **Discriminative Tasks:** For Screening, Disorder Classification, and Severity Assessment, we report the standard **Accuracy**. To account for the stochastic nature of LLM generation, we report the mean accuracy alongside the variance derived from 5 independent runs.
- **Generative Task:** For the Reasoning task, we assess the quality of generated explanations using both lexical and semantic metrics. We utilize **BLEU-1/4** [47] and **ROUGE-L** [37] to measure n -gram overlap with expert references. Furthermore, to evaluate semantic consistency beyond surface-level matching, we employ **BERTScore** [72] (based on the roberta-large [39] backbone), which correlates robustly with human judgment.

Computational Environment and Evaluation Pipeline. For open-source baselines, inference was conducted on a cluster equipped with 8 NVIDIA A100 GPUs. Closed-source models (GPT-4o and GPT-5) were accessed via their official APIs. All experiments utilized consistent hyperparameters (temperature=0.7, max tokens=2048, top-p=0.9) to ensure rigorous reproducibility.

To systematically address **RQ1**, we designed a two-phase evaluation protocol covering both broad-spectrum performance and counterfactual stability:

- (1) **Standard Performance Evaluation:** We primarily benchmarked the selected LLMs across the four mental health tasks using the **MHP-Eval-Base** dataset. To provide a rigorous comparative baseline for the discriminative tasks, we additionally evaluated specialized deep learning models, including **Mental-BERT** and **Mental-RoBERTa** [30], as well as **Bio-ClinicalBERT** [36] and **ClinicalBERT** [27]. To account for variance, five independent runs were performed for every model, with mean performance reported.
- (2) **Counterfactual Robustness Analysis:** To investigate the second aspect of **RQ1**—specifically how robust LLM capabilities are against semantic perturbations—we extended the evaluation to the **MHP-Eval-CF** dataset. This phase focuses exclusively on the generative models to determine their resilience when facing counterfactual input alterations.

4.2 Overall Performance Across the Task Spectrum

To address the performance spectrum of **RQ1**, we conducted a comprehensive evaluation of three categories of Large Language Models: Open-source General LLMs, Closed-source LLMs, and Open-source Mental-Health LLMs. Additionally, to establish a comparative baseline for discriminative capabilities, we incorporated Traditional PLMs (e.g., Mental-RoBERTa [30] and Mental-BERT [30]) specifically optimized for the mental health domain. Table 2 presents the detailed performance metrics, while Figure 4 visualizes the landscape across different tasks and paradigms. Our analysis reveals the following findings:

RQ1 Finding 1: LLMs Show No Advantage Over Traditional PLMs in Discriminative Tasks. LLMs do not yet surpass encoder-based discriminative PLMs on the two most decision-boundary-dominated tasks, *Screening* and *Severity*

Table 2. Performance comparison across four mental health tasks. Task headers *Screening*, *Disorder*, and *Severity* denote Screening, Disorder Classification, and Severity Assessment, respectively. For these tasks, Accuracy is reported, with variance displayed in subscripts. For the Reasoning task, evaluation metrics include BLEU-1 (B1), BLEU-4 (B4), ROUGE-L (R-L), and BERTScore (BS). Entries marked with “–” indicate that the model is incompatible with the specific task (not evaluated). The best and second-best results are highlighted in **bold** and underlined.

Model	Screening	Disorder	Severity	Reasoning			
	Acc.	Acc.	Acc.	B1	B4	R-L	BS
<i>Open-source General LLMs</i>							
Gemma-3-1B	0.5336 ($\pm 2.88 \times 10^{-7}$)	0.4674 ($\pm 0.00 \times 10^{-12}$)	0.3201 ($\pm 0.00 \times 10^{-12}$)	31.23	3.79	18.66	87.93
Llama-3.1-8B	0.5912 ($\pm 0.00 \times 10^{-12}$)	0.6380 ($\pm 0.00 \times 10^{-12}$)	0.3437 ($\pm 0.00 \times 10^{-12}$)	27.09	3.41	17.76	86.72
Mistral-7B-v0.3	0.5408 ($\pm 0.00 \times 10^{-12}$)	0.6247 ($\pm 6.65 \times 10^{-6}$)	0.3657 ($\pm 4.76 \times 10^{-6}$)	27.04	3.13	18.09	87.43
Phi-3-mini-4k(4B)	0.6264 ($\pm 0.00 \times 10^{-12}$)	0.6244 ($\pm 7.88 \times 10^{-5}$)	0.3151 ($\pm 3.26 \times 10^{-5}$)	28.77	2.89	17.71	87.52
Qwen2.5-1.5B	0.5800 ($\pm 2.03 \times 10^{-5}$)	0.5570 ($\pm 1.49 \times 10^{-5}$)	0.2701 ($\pm 4.88 \times 10^{-5}$)	26.99	3.30	18.48	87.70
Qwen2.5-3B	0.6404 ($\pm 6.93 \times 10^{-6}$)	0.5858 ($\pm 6.04 \times 10^{-6}$)	0.3553 ($\pm 7.89 \times 10^{-7}$)	29.23	3.48	19.10	87.73
Qwen2.5-7B	0.6028 ($\pm 1.87 \times 10^{-6}$)	0.6556 ($\pm 0.00 \times 10^{-12}$)	0.3496 ($\pm 0.00 \times 10^{-12}$)	29.23	3.48	19.10	87.97
Qwen3-4B	0.6188 ($\pm 1.73 \times 10^{-6}$)	0.6307 ($\pm 2.99 \times 10^{-6}$)	0.3239 ($\pm 5.01 \times 10^{-5}$)	27.42	2.41	16.28	86.87
Qwen3-8B	0.6384 ($\pm 7.52 \times 10^{-7}$)	0.6436 ($\pm 5.23 \times 10^{-6}$)	0.3212 ($\pm 4.32 \times 10^{-6}$)	28.52	2.90	17.75	87.74
Vicuna-7B-v1.5	0.5836 ($\pm 0.00 \times 10^{-12}$)	0.5374 ($\pm 6.60 \times 10^{-5}$)	0.2745 ($\pm 7.20 \times 10^{-5}$)	27.96	3.98	19.29	87.66
<i>Closed-source LLMs</i>							
GPT4o	0.6244 ($\pm 9.80 \times 10^{-7}$)	0.7184 ($\pm 3.20 \times 10^{-7}$)	0.3672 ($\pm 1.30 \times 10^{-6}$)	27.43	2.52	16.63	87.07
GPT5	0.6532 ($\pm 8.60 \times 10^{-7}$)	<u>0.7047</u> ($\pm 2.66 \times 10^{-5}$)	0.3983 ($\pm 1.10 \times 10^{-6}$)	27.44	2.51	16.55	87.01
<i>Open-source Mental-Health LLMs</i>							
Mental-LLM [65]	0.6372 ($\pm 1.54 \times 10^{-32}$)	–	0.4496 ($\pm 1.09 \times 10^{-5}$)	–	–	–	–
MentalLLaMA-13B [67]	0.5584 ($\pm 1.34 \times 10^{-4}$)	0.5310 ($\pm 3.68 \times 10^{-5}$)	0.2359 ($\pm 8.54 \times 10^{-5}$)	31.73	6.89	23.52	88.58
MentalLLaMA-7B [67]	0.6082 ($\pm 0.00 \times 10^{-12}$)	0.5138 ($\pm 4.30 \times 10^{-5}$)	0.2361 ($\pm 1.77 \times 10^{-6}$)	34.24	8.76	25.45	88.96
<i>Traditional Pre-trained Language Models in Mental Health</i>							
Mental-Roberta [30]	0.7372 ($\pm 3.95 \times 10^{-3}$)	0.6808 ($\pm 7.36 \times 10^{-3}$)	0.5000 ($\pm 1.00 \times 10^{-5}$)	–	–	–	–
Mental-Bert [30]	0.7324 ($\pm 2.90 \times 10^{-4}$)	0.6652 ($\pm 3.14 \times 10^{-3}$)	0.5346 ($\pm 1.10 \times 10^{-5}$)	–	–	–	–
Bio-ClinicalBERT [36]	0.6348 ($\pm 3.09 \times 10^{-3}$)	0.6272 ($\pm 1.59 \times 10^{-3}$)	0.3077 ($\pm 7.80 \times 10^{-5}$)	–	–	–	–
ClinicalBERT [27]	0.6296 ($\pm 2.27 \times 10^{-3}$)	0.3896 ($\pm 0.00 \times 10^{-12}$)	0.3077 ($\pm 1.80 \times 10^{-5}$)	–	–	–	–

Assessment. As illustrated in Figure 4 (Right), Traditional PLMs outperform the LLM average by roughly +0.08 accuracy on both Screening (0.684 vs. 0.603) and Severity (0.413 vs. 0.329), while reaching near parity on Disorder Classification (0.591 vs. 0.602). Importantly, this gap is not driven by a few outliers but remains consistent across models: *none* of the 15 evaluated LLMs exceeds the PLM mean on Screening, and only 1/15 does so on Severity (the fine-tuned Mental-LLM [65] at 0.4496), whereas the best PLMs (Mental-RoBERTa [30], Mental-BERT [30]) still form the top tier on these discriminative benchmarks (Table 2).

A plausible explanation is that these tasks primarily reward stable representation learning and sharp decision boundaries, where encoder-only architectures are well matched [26]; in contrast, prompt-based generative classification is known to be sensitive to answer priors, label verbalizations, and prompt formatting, which can prevent latent recognition ability from being expressed as a reliably calibrated label decision [73]. For mental health sensing systems, this suggests that specialized discriminative backbones remain strong candidates for front-end screening and severity scoring, while LLMs may be better positioned as complementary modules for downstream explanation or reasoning. Importantly, LLMs are not uniformly weaker: on Disorder Classification, they have largely caught up and can even set the performance ceiling (GPT-4o reaches 0.7184, exceeding the best PLM at 0.6808; 8/14 LLMs outperform the PLM mean; Table 2). Taken together, our results suggest that today’s LLMs are becoming competitive for disorder identification, yet remain unreliable for high-precision screening and severity quantification. This gap raises a key

question: *do LLMs lack mental-health knowledge, or do they possess it but fail to express it robustly under discriminative prompting?* (as the pivotal focus of **RQ2**; analysis and results in Sec.5)

RQ1 Finding 2: LLM Performance Diverges Across Tasks with a Notable Deficit in Severity Quantification. Overall, current LLMs are not yet consistently strong across the full spectrum of mental health sensing tasks, and their weakest point is *Severity Quantification*, where performance is uniformly poor across model families. Table 2 shows that LLMs can reach their best discriminative accuracy on Disorder Classification (up to 0.7184), while Severity Assessment stays markedly lower: most LLMs remain below 0.40, and even the best LLM result only reaches 0.4496. This pattern is stable rather than model-specific—despite differences in model scale and domain tuning, severity prediction remains the most error-prone task, suggesting that converting nuanced symptom evidence into a calibrated ordinal level is still a major bottleneck for today’s instruction-following LLMs.

Our error analysis further indicates that the severity failures are often *near-misses* with a systematic bias: among the incorrect severity predictions, 67.2% fall within ± 1 of the gold label, implying that LLMs frequently capture the rough severity neighborhood but fail to land on the correct intensity level. More concerning, 66.20% of the errors are *overestimates*, meaning the model tends to predict a more severe condition than warranted. This combination—high near-miss rate plus upward skew—suggests the limitation is less about recognizing symptom presence, and more about *intensity calibration and threshold placement*. In practical mental health sensing, such overestimation can inflate false alarms and undermine trust, highlighting the need for calibration-aware elicitation or complementary discriminative components when severity levels are required.

RQ1 Finding 3: Closed-source LLMs Define the Performance Ceiling while Small Open-source LLMs Show High Efficiency. Closed-source LLMs set the upper bound on overall discriminative performance, yet strong results are not exclusive to large parameter scales. On average across the three discriminative tasks, Closed-source LLMs outperform Open-source General LLMs by a clear margin (Screening: 0.639 vs. 0.595, +0.043; Disorder Classification: 0.712 vs. 0.596, +0.115;

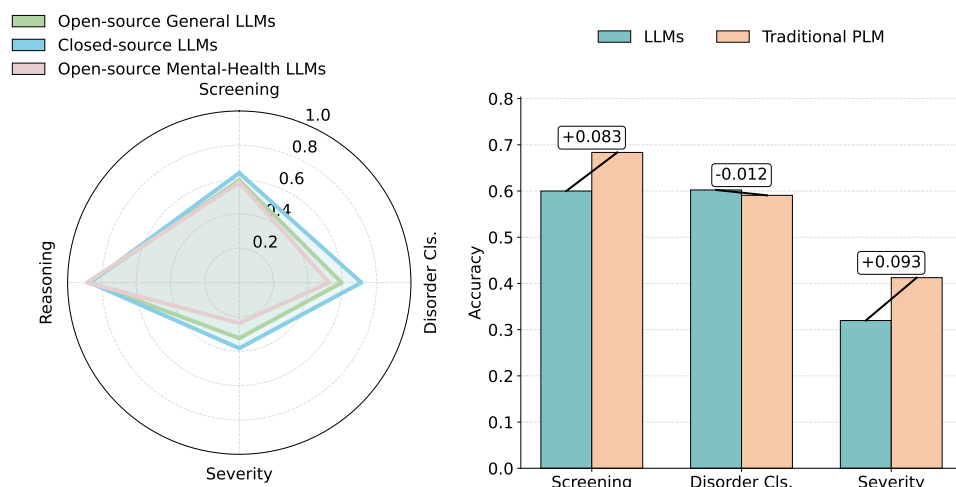


Fig. 4. **(Left)** A radar chart visualizing the average performance of three LLM categories (Open-source General, Closed-source, and Mental-Health LLMs) across the four sensing tasks. **(Right)** A comparative analysis between the aggregated average of LLMs and Traditional Pre-trained Language Models (PLMs) on the three discriminative tasks.

Severity Assessment: 0.383 vs. 0.324, +0.059), indicating a consistent ceiling effect beyond individual model variance. This ceiling is most pronounced for Disorder Classification: GPT-4o reaches 0.7184 (Table 2), exceeding the best Open-source General LLM (Qwen2.5-7B [49] at 0.6556) by +0.0628, suggesting that closed-source models currently hold a notable advantage in capturing and expressing disorder-type knowledge under instruction-following settings. Meanwhile, several small open-source LLMs achieve a favorable accuracy–efficiency trade-off. For example, Qwen2.5-3B [49] attains a comparable three-task macro-average to Llama-3.1-8B [22] (0.527 vs. 0.524), and even surpasses it on Screening (0.6404 vs. 0.5912, +0.0492), highlighting that scale alone does not reliably predict clinical discriminative capability.

RQ1 Finding 4: Mental-Health Fine-tuning Boosts In-Task Performance, but Cross-Task Adaptation Is Unreliable. Domain alignment helps when the evaluation protocol matches the fine-tuning schema, yet the gains do not reliably transfer once the task format or label space shifts. We observe clear *in-task* benefits from mental-health tuning: for instance, Mental-LLM [65] performs strongly on the two discriminative settings it targets (Screening: 0.6372, Severity: 0.4496), exceeding the open-source general LLM average (Screening: ≈ 0.596 , Severity: ≈ 0.324). However, this advantage is not stable beyond its trained schemas: in our evaluation, Mental-LLM [65] often fails to follow the required output space for other formats (e.g., producing Screening/Severity-style responses when prompted for Disorder Classification or open-ended Reasoning, showcase in Appendix A.1), and it shows pronounced robustness degradation under counterfactual perturbations (e.g., Severity drops by -0.271 in Table 2). A complementary pattern appears in MentaLLaMA [67]. According to its paper, it is fine-tuned on IMHI-style objectives that emphasize (i) binary detection, (ii) multi-class disorder detection, and (iii) reasoning generation—which aligns with its relatively strong reasoning-oriented score in our benchmark (BS: 88.96/88.58, about +1.2 above the open-source general mean). Yet this alignment does *not* translate into better severity quantification (Severity: 0.236 for both 7B/13B, roughly -0.09 below the open-source general mean). Our results suggest that “mental-health ability” learned via instruction SFT is often task-schema bound and may not transfer reliably across task formats. This raises a natural question: when models succeed on detection-style tasks, are they drawing on stable mental-health knowledge, or simply replaying schema-conditioned response patterns—a research question described in RQ2.

4.3 Robustness Under Counterfactual Perturbations

To examine robustness beyond standard accuracy, we evaluate models on MHP-Eval-CF, where each instance is paired with a counterfactually perturbed variant and the performance shift is measured as $\Delta = \text{CF} - \text{Base}$. Table 3 reports Base/CF accuracies and Δ across Screening, Disorder Classification (Disorder Cls.), and Severity Assessment (Severity), while Figure 5 provides complementary visual summaries (model-level Δ heatmap, group-level mean $\pm$ SD, and an accuracy–robustness scatter). Overall, counterfactual perturbations consistently reduce performance across model families, revealing robustness gaps that are not apparent from Base accuracy alone.

RQ1 Robustness Finding 1: Counterfactual Perturbations Consistently Degrade Discriminative Performance, with Disorder Classification Being the Most Fragile. Across all evaluated models and tasks, Δ is uniformly negative, indicating that counterfactual perturbations systematically hurt discriminative accuracy rather than producing mixed gains and losses. At the task level, Disorder Cls. shows the largest average degradation (mean $\Delta \approx -0.1145$, median $\Delta \approx -0.1170$), followed by Screening (mean ≈ -0.0968), while Severity exhibits a smaller typical drop (median ≈ -0.0540) but with occasional extreme failures (Table 3). This pattern is also stable within the open-source general LLMs: all models suffer sizeable drops on Disorder Cls. (ranging roughly from -0.0765 to -0.1537), suggesting a broad sensitivity to counterfactual edits rather than an idiosyncratic weakness of a few models. A plausible explanation is that disorder

Table 3. Comparative results on MHP-Eval-CF across Screening, Disorder Classification (Disorder Cls.), and Severity Assessment (Severity). Accuracy is presented for both the original (**Base**) and counterfactual (**CF**) settings. The performance shift is quantified as $\Delta = \text{CF} - \text{Base}$, where negative values (marked in red) signify performance degradation. Within the Δ columns, the **largest perturbations** (most negative values) are highlighted in bold, while the smallest perturbations are underlined.

Model	Screening			Disorder Cls.			Severity		
	Base	CF	Δ	Base	CF	Δ	Base	CF	Δ
<i>Open-source General LLMs</i>									
Gemma-3-1B	0.5336	0.4492	↓ -0.0844	0.4674	0.3577	↓ -0.1097	0.3201	0.2705	↓ -0.0496
Llama-3.1-8B	0.5912	0.5221	↓ -0.0691	0.6380	0.5053	↓ -0.1327	0.3437	0.2685	↓ -0.0752
Mistral-7B	0.5408	0.4349	↓ -0.1059	0.6247	0.5183	↓ -0.1064	0.3657	0.2723	↓ -0.0934
Phi-3-mini-4k(4B)	0.6264	0.5015	↓ -0.1249	0.6244	0.5218	↓ -0.1026	0.3151	0.3009	↓ -0.0142
Qwen2.5-1.5B	0.5800	0.4944	↓ -0.0856	0.5570	0.4805	↓ -0.0765	0.2701	0.2161	↓ -0.0540
Qwen2.5-3B	0.6404	0.5179	↓ -0.1225	0.5858	0.4616	↓ -0.1242	0.3553	0.2813	↓ -0.0741
Qwen2.5-7B	0.6028	0.5036	↓ -0.0992	0.6556	0.5266	↓ -0.1290	0.3496	0.2765	↓ -0.0731
Qwen3-4B	0.6188	0.4974	↓ -0.1214	0.6307	0.5313	↓ -0.0994	0.3239	0.2652	↓ -0.0587
Qwen3-8B	0.6384	0.5262	↓ -0.1122	0.6436	0.5089	↓ -0.1347	0.3212	0.2756	↓ -0.0456
Vicuna-7B	0.5836	0.4790	↓ -0.1046	0.5374	0.3837	↓ -0.1537	0.2745	0.2054	↓ -0.0691
<i>Closed-source LLMs</i>									
GPT4o	0.6244	0.6134	↓ -0.0110	0.7184	0.6860	↓ -0.0324	0.3672	0.3499	↓ -0.0173
GPT5	0.6532	0.5732	↓ -0.0800	0.7047	0.6923	↓ -0.0124	0.3983	0.3847	↓ -0.0136
<i>Open-source Mental-Health LLMs</i>									
Mental-LLM	0.6372	0.5662	↓ -0.0710	–	–	–	0.4496	0.1786	↓ -0.2710
MentalLaMA-13B	0.5584	0.4144	↓ -0.1440	0.5310	0.3495	↓ -0.1815	0.2359	0.2038	↓ -0.0322
MentalLaMA-7B	0.6082	0.4923	↓ -0.1159	0.5138	0.3065	↓ -0.2073	0.2361	0.2121	↓ -0.0240

identification relies on relatively sparse, high-impact symptom cues and label-bound lexical patterns; counterfactual perturbations that alter these cues can easily shift model decisions even when the clinical intent remains close, making Disorder Cls. particularly brittle. For mental health sensing, this finding emphasizes that high Base accuracy does not imply reliable behavior under realistic textual variations, and robustness should be treated as a first-class evaluation axis rather than an afterthought.

RQ1 Robustness Finding 2: Closed-source LLMs Are Substantially More Robust on Average, but Robustness Is Still Task-Dependent. Robustness differs sharply by model family. Closed-source LLMs show markedly smaller perturbation sensitivity overall (mean $\Delta_{\text{avg}} \approx -0.0279$) compared to open-source general LLMs (mean $\Delta_{\text{avg}} \approx -0.0941$) and open-source mental-health LLMs (mean $\Delta_{\text{avg}} \approx -0.1350$), consistent with the separation visible in Figure 5 (middle). Notably, the smallest degradations in the table come from closed-source systems on specific tasks (e.g., GPT-4o on Screening: -0.0110 ; GPT-5 on Disorder Cls.: -0.0124 ; GPT-5 on Severity: -0.0136), suggesting stronger instruction adherence and more stable decision behavior under perturbation. However, robustness is not uniformly guaranteed even for the best family: GPT-5 still shows a non-trivial drop on Screening (-0.0800), indicating that robustness depends on the task and the perturbation type rather than being a single global property. Practically, this implies that robustness improvements should be validated per task, and closed-source models can serve as useful upper-bound references for robustness when benchmarking mental health sensing systems.

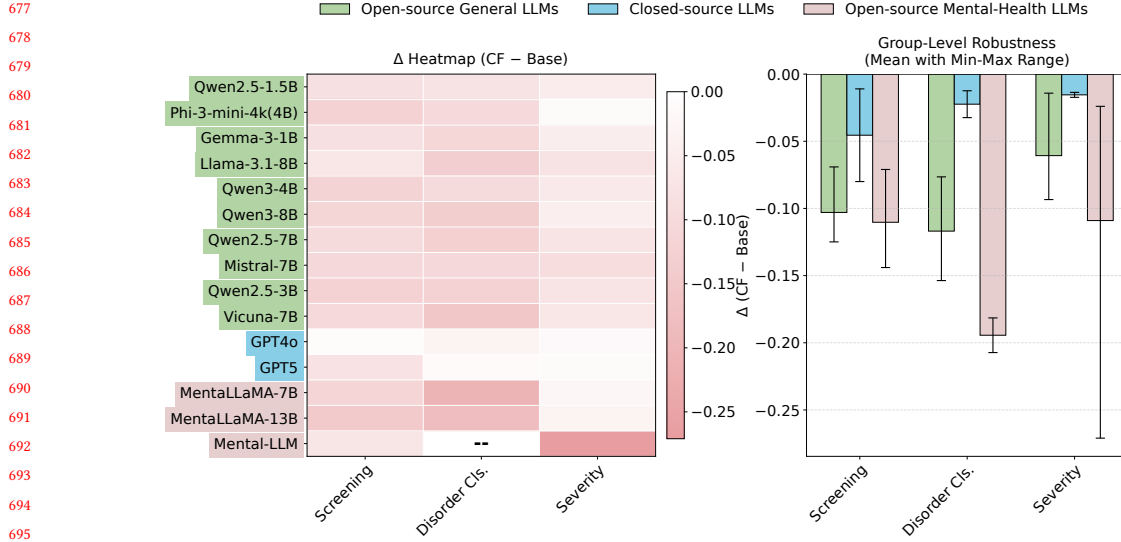


Fig. 5. Counterfactual robustness analysis on MHP-Eval-CF. **(Left)** Model-level heatmap of performance shifts $\Delta = \text{CF} - \text{Base}$ across tasks, where darker cells indicate larger degradations. **(Right)** Group-level robustness summarized as the **mean** of Δ for each task, with error bars representing the **Min-Max range** (indicating the spread from the smallest to the largest performance degradation within each group).

RQ1 Robustness Finding 3: Domain-Aligned Mental-Health LLMs Are Not Necessarily More Robust and Can Exhibit Catastrophic Failures Under Perturbations. Domain alignment does not translate into robustness by default. In fact, the largest robustness failures in Table 3 occur in the mental-health LLM group, with extreme drops such as Mental-LLM on Severity ($0.4496 \rightarrow 0.1786$, $\Delta = -0.2710$) and large Disorder Cls. degradations for MentaLLaMA variants (e.g., MentaLLaMA-7B: $\Delta = -0.2073$; MentaLLaMA-13B: $\Delta = -0.1815$). These failures are not merely small fluctuations around a decision boundary; they indicate that counterfactual edits can trigger pronounced shifts in predicted severity or diagnosis, consistent with over-reliance on schema- or surface-form cues learned during domain tuning. Moreover, Figure 5 (right) suggests that robustness is only weakly coupled with Base accuracy: some models with competitive Base accuracy remain brittle, while others with moderate Base accuracy show comparatively smaller drops on certain tasks (e.g., the small Δ on Severity for Phi-3-mini-4k(4B), $\Delta = -0.0142$). For mental health sensing, this finding highlights a key risk: domain fine-tuning may improve in-distribution accuracy yet still yield fragile behavior under clinically plausible perturbations, motivating robustness-aware training/evaluation and careful deployment constraints when text inputs can vary substantially in expression style.

5 RQ2: Measuring LLM Knowledge on Mental Health

To evaluate the reliability of LLMs in mental health sensing, relying solely on the generated output provides an incomplete assessment. An incorrect response or a hallucination does not necessarily imply that the model lacks the relevant mental health knowledge within its parameters. Recent research has demonstrated that models often encode truthful information in their latent states even when their external outputs are incorrect [3, 7, 42, 46]. Therefore, it is essential to distinguish between what the model knows and what it generates.

5.1 Definitions of External and Internal Knowledge

Drawing on the frameworks established by Gekhman et al. [21] and Orgad et al. [46], we propose a systematic method to measure LLM knowledge on the MHP-Eval task. We employ a cognitive analogy to conceptualize this framework: Internal Knowledge represents the “thought” process, which is encoded in the deep neural representations, while External Knowledge represents “speech”, which is the final text produced after the decoding process.

We formally define knowledge as the capability of the model to distinguish the correct answer from a set of plausible but incorrect answers. Given a mental health question q , a correct answer a , and a set of incorrect options $\tilde{A}(q)$, we calculate a knowledge score K_q using a scoring function S . The score is defined as the proportion of pairs where the correct answer is ranked higher than the incorrect one:

$$K_q(S) = \frac{1}{|\Omega|} \sum_{(a, \tilde{a}) \in \Omega} \mathbb{I}(S(q, a) > S(q, \tilde{a})) \quad (1)$$

where Ω denotes the set of all pairs comprising the correct answer and an incorrect answer, and $\mathbb{I}$ is the indicator function. Based on this formulation, we define two distinct scoring functions to measure external and internal knowledge.

First, the **External Knowledge Score** (S_{ext}) measures the model’s explicit confidence in a generated answer. Given the autoregressive nature of LLMs, the joint probability of an answer a conditioned on a question q is modeled as the product of conditional token probabilities: $P_{LM}(a|q) = \prod_{i=1}^{|a|} P(a_i|q, a_{<i})$, where a_i denotes the i -th token of the answer sequence and $a_{<i}$ represents the preceding context tokens. Since this raw joint probability naturally diminishes with sequence length, introducing a bias against longer responses, we adopt length-normalized log-probability to ensure fair comparison [61]. The final score is derived by transforming the multiplicative probability into an averaged log-likelihood:

$$\begin{aligned} S_{ext}(q, a) &= P_{LM}(a|q)_{\text{norm}} = \left(\prod_{i=1}^{|a|} P(a_i|q, a_{<i}) \right)^{\frac{1}{|a|}} \\ &= \exp \left(\frac{1}{|a|} \sum_{i=1}^{|a|} \log P(a_i|q, a_{<i}) \right) \end{aligned} \quad (2)$$

Here, the exponential function (exp) maps the averaged log-likelihood back to the probability space $[0, 1]$, mathematically corresponding to the geometric mean of the token probabilities.

Second, the **Internal Knowledge Score** (S_{int}) reflects the model’s latent confidence, derived using a linear probing classifier. This classifier operates on the intermediate representations of the model to predict the semantic category of the input. Specifically, we implement the probe using a multi-class logistic regression model, which maps the high-dimensional hidden states to a probability distribution over the candidate class set $\mathcal{Y}$. Since knowledge may be distributed across different depths of the network, we train a separate probe for each layer $l \in \{1, \dots, L\}$. To ensure statistical robustness and mitigate overfitting, we conduct the training using a 5-fold cross-validation scheme. The internal score is defined as the probability assigned to the target class label $y_a \in \mathcal{Y}$ by the probe at the optimal layer l^* , which is selected based on the highest cross-validation accuracy:

$$S_{int}(q, a) = P(y = y_a | \mathbf{h}^{(l^*)}(q); \theta^{(l^*)}) \quad (3)$$

where $\theta^{(l^*)}$ denotes the learned parameters of the probe at the optimal layer. This metric captures the model’s true capability to discriminate among categories internally, independent of external generation constraints.

5.2 Overall Analysis: The Knowing but Not Saying Phenomenon

To conduct a rigorous quantitative analysis, we prioritize tasks with objective ground truth labels that permit unambiguous verification. Accordingly, our evaluation focuses on three representative tasks from the MHP-Eval dataset: Screening, Disorder Classification, and Severity Assessment. To offer a holistic view of model capabilities across these diverse tasks, we define an aggregated knowledge metric. For a given model, the overall internal (K_{int}^{avg}) and external (K_{ext}^{avg}) knowledge scores are calculated as the arithmetic mean across the task set $\mathcal{T} = \{\text{Screening, Disorder Cls., Severity}\}$:

$$K_*^{avg} = \frac{1}{|\mathcal{T}|} \sum_{t \in \mathcal{T}} K_*(t) \quad (4)$$

where $*$ $\in \{int, ext\}$ denotes the knowledge type.

Aggregated Knowledge Gap Analysis. Our comparative analysis reveals a pervasive and significant trend: the internal knowledge scores consistently surpass the external knowledge scores for every model evaluated. As visualized in Figure 6, this discrepancy is systematic across all 10 open-source LLMs. Quantitatively, we observe a substantial average knowledge gap of 0.2976, indicating that nearly third of the model’s internal medical insights are lost during the explicit generation process (next token prediction). This performance gap demonstrates that while models successfully encode truthful mental health information within their latent representations (high K_{int}^{avg}), they fail to faithfully manifest this understanding in their explicit probabilistic outputs (low K_{ext}^{avg}). This observation aligns with findings in general domain studies [46], confirming that the sub-optimal performance of LLMs in mental health sensing often stems from a *misalignment between latent representation and token generation*, rather than a fundamental lack of medical knowledge. Consequently, we formally term this structural discrepancy the “**Knowing but Not Saying**” phenomenon.

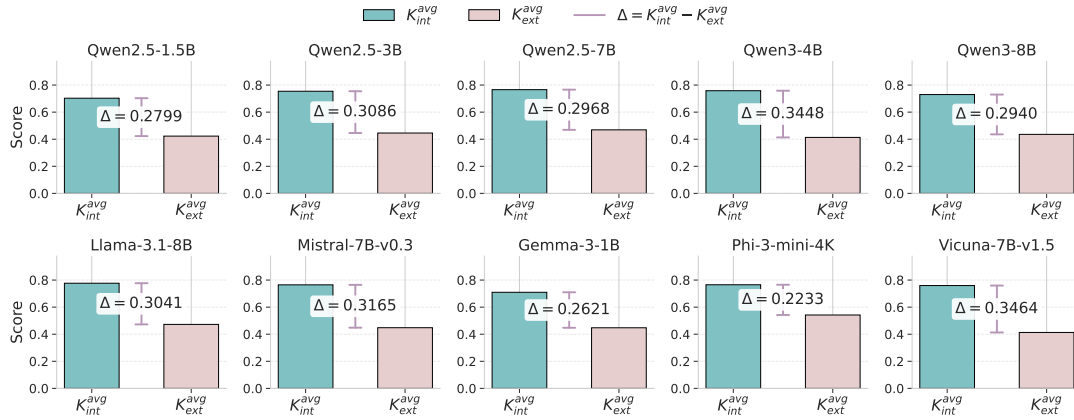


Fig. 6. Aggregated Internal vs. External Knowledge Scores across 10 open-source LLMs. The results are averaged over three tasks: Screening, Disorder Classification, and Severity Assessment. The K_{int}^{avg} and K_{ext}^{avg} denote the aggregated internal and external scores, respectively. The consistent gap (Δ) highlights the model’s unexpressed latent knowledge.

Task-Level Analysis. A critical insight emerges from our fine-grained analysis: the knowledge gap (Δ) expands significantly when transitioning from simple binary recognition to complex diagnostic reasoning. As detailed in Table 4, the simpler “Screening” task exhibits the narrowest mean gap of 0.2374. However, as task complexity escalates to

Table 4. Comparison of Internal Knowledge (K_{int}) and External Knowledge (K_{ext}) across three mental health tasks. The gap $\Delta = K_{int} - K_{ext}$ measures the discrepancy between latent representation and token generation. Darker shades of red indicate a larger gap, signifying that the model struggles more severely to express its latent knowledge in that specific setting.

Model	Screening			Disorder Classification			Severity Assessment		
	K_{int}	K_{ext}	Δ	K_{int}	K_{ext}	Δ	K_{int}	K_{ext}	Δ
Qwen2.5-1.5B	0.8140	0.5964	0.2176	0.7696	0.4844	0.2852	0.5250	0.1880	0.3370
Qwen2.5-3B	0.8188	0.6440	0.1748	0.7722	0.3936	0.3786	0.6720	0.2996	0.3724
Qwen2.5-7B	0.8268	0.6092	0.2176	0.7726	0.4520	0.3206	0.6985	0.3464	0.3521
Qwen3-4B	0.8356	0.5468	0.2888	0.7867	0.4148	0.3719	0.6522	0.2785	0.3738
Qwen3-8B	0.8250	0.5600	0.2650	0.8070	0.4156	0.3914	0.5575	0.3320	0.2255
Llama-3.1-8B	0.8281	0.6540	0.1741	0.8109	0.4396	0.3713	0.6910	0.3240	0.3670
Gemma-3-1B	0.7982	0.5596	0.2386	0.7339	0.4844	0.2495	0.5970	0.2988	0.2982
Mistral-7B-V0.3	0.8197	0.5380	0.2817	0.8052	0.3880	0.4172	0.6690	0.4184	0.2506
Vicuna-7B-V1.5	0.8320	0.5076	0.3244	0.7682	0.4096	0.3586	0.6780	0.3220	0.3560
Phi-3-Mini-4K(4B)	0.8281	0.6364	0.1917	0.7777	0.6348	0.1429	0.6900	0.3548	0.3352
Mean	0.8226	0.5852	0.2374	0.7804	0.4517	0.3287	0.6430	0.3163	0.3268

multi-class “Disorder Classification” and fine-grained “Severity Assessment,” the gap widens and stabilizes at 0.3287 and 0.3268, respectively. Crucially, this persistent divergence is driven by a severe deterioration in external expression (K_{ext} plummets from 0.5852 to 0.3163). While internal knowledge also reflects the increasing difficulty (dropping to 0.6430 in Severity Assessment), it consistently maintains a dominant lead over explicit output. This indicates that *the expression bottleneck remains a persistent barrier regardless of task complexity*. Even when the model’s internal certainty wavers in the most challenging scenarios, its ability to articulate the answer degrades far more rapidly than its latent understanding. This persistent gap implies that the core capability is already present but suppressed, highlighting that the priority lies not in retraining, but in developing effective methods to unlock and externalize this hidden potential.

5.3 Layer-wise Analysis: The Peak of “Knowing” in Intermediate Depth

Our fine-grained probing analysis reveals a pivotal structural phenomenon: the most accurate mental health representations consistently reside in the intermediate-to-deep layers, rather than the final output layer. As visualized in Figure 7, we track the trajectory of internal knowledge (K_{int}) across the complete depth of LLMs. The heatmaps exhibit a consistent trend where representational quality maximizes in the 50% – 80% depth range, followed by a sharp deterioration at the terminal layers. Crucially, this late-stage degradation is particularly pronounced in the complex “Disorder Classification” task. This structural evidence reinforces the existence of an *expression bottleneck*: while models successfully encode rich clinical insights in their latent space, this information is partially diluted or distorted during the transition to the final next-token prediction. This observation underscores a pressing priority: to develop mechanisms that ensure the faithful propagation of these optimal intermediate representations to the final output space, thereby bridging the gap between robust internal understanding and accurate external expression.

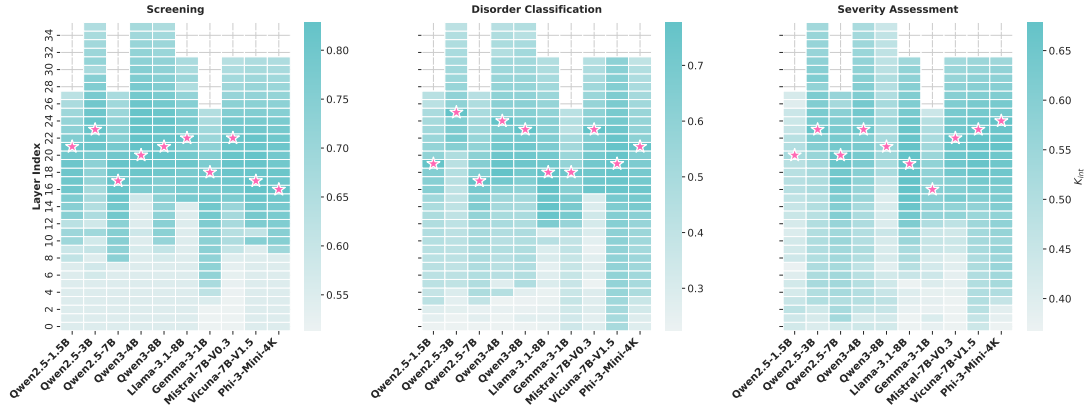


Fig. 7. Visualization of Layer-wise Internal Knowledge (K_{int}). The heatmaps display the probing accuracy across all transformer layers for 10 LLMs on three mental health tasks: Screening, Disorder Classification, and Severity Assessment. The color intensity represents the magnitude of K_{int} , and the pink stars (*) indicate the specific layer achieving the maximum K_{int} for each model.

6 RQ3: Mental-aware Knowledge Enhancement (MaKE) Framework

While existing Large Language Models (LLMs) demonstrate latent capabilities in identifying mental health indicators (as discussed in RQ1), their explicit generation is often hampered by safety alignment and generation biases. To bridge this gap, we propose the **Mental-aware Knowledge Enhancement (MaKE)** framework. MaKE is a training-free, inference-time intervention mechanism that unlocks and utilizes the model’s internal “mental wisdom.”

As illustrated in Figure 8, the framework operates in three sequential stages: (1) *Mental Knowledge Probing* to localize optimal semantic representations; (2) *Internal Mental Knowledge Construction* to build a robust, non-parametric reference system; and (3) *Inference Intervention* to rectify hallucinations dynamically.

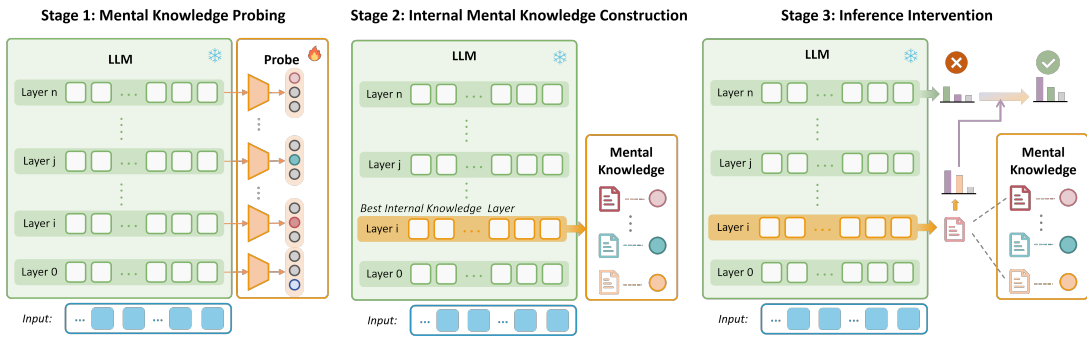


Fig. 8. **Overview of the MaKE framework.** The pipeline consists of three stages: (1) **Mental Knowledge Probing**, identifying the optimal intermediate layer rich in mental health semantics via probing; (2) **Internal Mental Knowledge Construction**, building a non-parametric repository of calibrated latent representations from the localized layer; and (3) **Inference Intervention**, correcting model hallucinations during inference by retrieving nearest semantic exemplars and applying a mental-aware hard switch.

6.1 MaKE Framework

Based on the metrics defined in Section 5.1, we propose the **Mental-aware Knowledge Enhancement (MaKE)** framework. MaKE is a training-free, non-parametric mechanism designed to bridge the gap between the LLM’s internal latent capabilities and its external generation. As shown in Figure 8, the framework operates in three streamlined phases: *Mental Knowledge Probing*, *Internal Mental Knowledge Construction*, and *Inference Intervention*.

Phase 1: Mental Knowledge Probing. The goal is to identify the specific layer where the model’s internal representation of mental health concepts is most discriminative. Utilizing the Internal Knowledge Score (S_{int}) from Section 5.1, we evaluate the linear separability of representations across all layers. We select the optimal layer L^* that maximizes the probing accuracy:

$$L^* = \operatorname{argmax}_{l \in \{1, \dots, L\}} \operatorname{Accuracy}(S_{int}^{(l)}) \quad (5)$$

Layer L^* serves as the semantic anchor, providing high-dimensional features that are robust against generation biases.

Phase 2: Internal Mental Knowledge Construction. We construct a mental knowledge bank ($\mathcal{M}$) to index the information contained in layer L^* . We extract the hidden states $\mathbf{h}_i^{(L^*)}$ for all available samples. To enhance semantic distinctiveness and remove common background noise, we apply a feature calibration operation. Specifically, we compute the global mean vector $\boldsymbol{\mu}$ and apply a centering transformation controlled by a factor β :

$$\hat{\mathbf{v}}_i = \mathbf{h}_i^{(L^*)} - \beta \cdot \boldsymbol{\mu} \quad (6)$$

The calibrated vectors are paired with their ground-truth labels to form the repository $\mathcal{M} = \{(\hat{\mathbf{v}}_i, y_i)\}_{i=1}^N$. This process effectively maps the model’s internal activations to explicit diagnostic criteria without parameter updates.

Phase 3: Inference Intervention. During inference, MaKE integrates the pre-computed components to calibrate the model’s prediction. For a given input x , we first apply **Contextual Centering** to its hidden representation to obtain a calibrated vector $\hat{\mathbf{v}}_x$. This vector is then used to query the datastore $\mathcal{M}$ via **Knowledge Retrieval**, identifying the k -nearest semantic exemplars, $\mathcal{N}_k(x)$, based on cosine similarity.

We aggregate the labels of these neighbors to form a retrieval-based distribution (P_{retr}), which reflects the consensus of the local manifold structure:

$$P_{retr}(c) = \frac{\exp(s_c)}{\sum_{c' \in \mathcal{Y}} \exp(s_{c'})}, \quad \text{where } s_c = \gamma \sum_{(\hat{\mathbf{v}}_j, y_j) \in \mathcal{N}_k(x)} \mathbb{I}(y_j = c) \cdot \cos(\hat{\mathbf{v}}_x, \hat{\mathbf{v}}_j) \quad (7)$$

Here, γ is a scaling factor. To determine the necessity of intervention, we quantify the conflict between the model’s original external output (P_{ext}) and the retrieval evidence (P_{retr}) using the cognitive dissonance score ($\mathcal{D}$):

$$\mathcal{D}(x) = \frac{1}{2} (D_{KL}(P_{ext} || P_{retr}) + D_{KL}(P_{retr} || P_{ext})) \quad (8)$$

Finally, we employ a dynamic gating strategy. If the dissonance $\mathcal{D}(x)$ exceeds a threshold τ —implying a high risk of hallucination where the generation contradicts robust internal knowledge—we override the parametric output with the non-parametric retrieval consensus; otherwise, we retain the model’s original prediction:

$$\hat{y} = \begin{cases} \operatorname{argmax} P_{retr} & \text{if } \mathcal{D}(x) > \tau \\ \operatorname{argmax} P_{ext} & \text{otherwise} \end{cases} \quad (9)$$

Model	Screening					Disorder Cls.					Severity				
	Base	ICL	RAG	SFT	Ours	Base	ICL	RAG	SFT	Ours	Base	ICL	RAG	SFT	Ours
Gemma-3-1B	0.5336	0.5353	0.5552	0.5360	0.8980	0.4674	0.4988	0.6315	0.6304	0.7116	0.3201	0.3391	0.3336	0.3121	0.3792
Llama-3.1-8B	0.5912	0.6919	0.5839	0.5864	0.9076	0.6380	0.6872	0.6454	0.6464	0.8424	0.3437	0.3399	0.3562	0.3317	0.7684
Mistral-7B-v0.3	0.5408	0.5833	0.5800	0.6840	0.9008	0.6247	0.6513	0.6368	0.7368	0.7480	0.3657	0.3781	0.4113	0.4081	0.5368
Phi-3-mini-4k(4B)	0.6264	0.6122	0.6792	0.6192	0.9236	0.6244	0.6241	0.5936	0.6549	0.8486	0.3151	0.3265	0.3355	0.3271	0.7676
Qwen2.5-1.5B	0.5800	0.6066	0.5688	0.6288	0.9088	0.5570	0.5504	0.5888	0.5918	0.6100	0.2701	0.2792	0.3747	0.3706	0.4740
Qwen2.5-3B	0.6404	0.7344	0.6752	0.6600	0.9052	0.5858	0.5958	0.6675	0.6668	0.6968	0.3553	0.3749	0.3626	0.3592	0.4352
Qwen2.5-7B	0.6028	0.6378	0.7120	0.6208	0.9014	0.6556	0.6744	0.6552	0.6656	0.7060	0.3496	0.3632	0.4273	0.3435	0.4644
Qwen3-4B	0.6188	0.6627	0.6176	0.6392	0.9148	0.6307	0.6737	0.6624	0.6397	0.7660	0.3239	0.3413	0.4209	0.3535	0.5125
Qwen3-8B	0.6384	0.6607	0.6288	0.6340	0.9072	0.6436	0.6704	0.4680	0.6664	0.8120	0.3212	0.3407	0.3201	0.3411	0.7120
Vicuna-7B-v1.5	0.5836	0.5933	0.5774	0.5720	0.9092	0.5374	0.5453	0.5654	0.5810	0.8424	0.2745	0.2617	0.3123	0.2886	0.7592
Average	0.5956	0.6318	0.6178	0.6180	0.9077	0.5965	0.6172	0.6115	0.6480	0.7584	0.3239	0.3344	0.3655	0.3435	0.5809

Table 5. Comparison results of MaKE (Ours) against In-Context Learning (ICL), Retrieval-Augmented Generation (RAG), and Supervised Fine-Tuning (SFT) across three mental health tasks. Higher is better.

This mechanism ensures that the final prediction is grounded in retrieved clinical evidence, specifically when the model exhibits uncertainty or bias.

6.2 Evaluation

6.2.1 Comparison to Common Adaptation Baselines. Table 5 compares MaKE with three common adaptation baselines (ICL, RAG, and SFT) on Screening, Disorder Classification, and Severity across ten open-weight LLM backbones. MaKE achieves the best average performance on all three tasks, improving from 0.5956 to 0.9077 on Screening, from 0.5965 to 0.7584 on Disorder Classification, and from 0.3239 to 0.5809 on Severity. In contrast, the strongest competing baselines are notably lower: for Screening, the best baseline reaches 0.6318 (ICL); for Disorder Classification, the best baseline reaches 0.6480 (SFT); and for Severity, the best baseline reaches 0.3655 (RAG).

The gains are consistent across model families and scales. For example, on Screening, MaKE boosts Gemma-3-1B from 0.5336 (baseline) to 0.8980 and improves Qwen2.5-7B from 0.6028 to 0.9014. On Disorder Classification, MaKE improves Llama-3.1-8B from 0.6380 to 0.8424 and raises Vicuna-7B-v1.5 from 0.5374 to 0.8424. On Severity, MaKE yields particularly large improvements for several 7–8B models, including Llama-3.1-8B (0.3437→0.7684) and Vicuna-7B-v1.5 (0.2745→0.7592). Overall, these results indicate that MaKE provides a strong and stable advantage over prompting-, retrieval-, and fine-tuning-based alternatives under the same evaluation setting.

6.2.2 Ablation Study. To investigate the individual contributions of MaKE’s core components, we conducted a comprehensive ablation study across the Screening, Disorder Classification, and Severity Assessment tasks. Table 6 reports the performance degradation (Δ) when selectively removing the **Knowledge Retrieval** (w/o Ret.) and **Contextual Centering** (w/o Cen.) modules. The results reveal distinct roles for each component.

Criticality of Knowledge Retrieval. The results unequivocally establish domain-specific retrieval as the cornerstone of the MaKE framework. As shown in the *w/o Ret.* columns, removing the retrieval module leads to catastrophic performance drops across all tasks and model architectures. In the **Screening** task, the absence of retrieved evidence causes an average accuracy drop of over 20% for most models (e.g., Gemma-3-1B drops by 27.04%). The impact is even more pronounced in the **Severity Assessment** task, where models require precise clinical criteria to distinguish

Table 6. Ablation study on the three sub-tasks. We report the Accuracy (%) of the full **MaKE** framework and the performance drop (Δ) when removing specific components. **w/o Ret.**: Removal of Retrieval. The most significant drops are highlighted in **bold**.

Model	Screening			Disorder Classification			Severity Assessment		
	MaKE	w/o Cen.	w/o Ret.	MaKE	w/o Cen.	w/o Ret.	MaKE	w/o Cen.	w/o Ret.
Gemma-3-1B	89.80	-0.16	-27.04	71.16	-20.20	-29.72	37.92	0.00	-12.84
Llama-3.1-8B	90.76	-1.16	-20.60	84.24	0.00	-16.20	76.84	0.00	-39.36
Mistral-7B	90.08	-0.04	-19.68	74.80	-10.48	-22.60	53.68	0.00	-11.04
Phi-3-mini-4k(4B)	92.36	-0.76	-22.40	84.86	-0.48	-16.79	76.76	0.00	-45.28
Qwen2.5-1.5B	90.88	-0.36	-25.56	61.00	-0.16	-8.24	47.40	-10.40	-21.80
Qwen2.5-3B	90.52	-0.48	-22.64	69.68	-30.00	-30.12	43.52	-10.00	-10.00
Qwen2.5-7B	90.14	-0.22	-21.34	70.60	-20.04	-20.60	46.44	-10.00	-10.00
Qwen3-4B	91.48	-0.16	-24.72	76.60	-30.00	-30.00	51.25	-20.00	-20.00
Qwen3-8B	90.72	-1.84	-24.52	81.20	-20.04	-28.72	71.20	-0.20	-29.00
Vicuna-7B	90.92	-0.28	-21.56	84.24	-0.16	-15.68	75.92	0.00	-48.36

intensity levels; here, removing retrieval renders models like Vicuna-7B ($\Delta = -48.36\%$) and Phi-3-mini ($\Delta = -45.28\%$) nearly ineffective. This confirms that while general LLMs possess linguistic capabilities, they fundamentally lack the specialized clinical knowledge required for mental health inference, which our retrieval mechanism effectively supplements.

Model-Dependent Correction via Centering. The impact of Contextual Centering varies significantly across model families, highlighting its role as an adaptive bias-correction mechanism. For models that are inherently well-calibrated, such as Llama-3.1-8B and Mistral-7B, removing centering has negligible impact (e.g., $\Delta \approx 0.00$ in Disorder Classification and Severity), indicating these models naturally utilize the retrieved context without significant prior bias. However, for the *Qwen series*, Centering proves critical. In *Disorder Classification*, removing this module causes performance to collapse by up to 30.00% for Qwen2.5-3B and Qwen3-4B. This suggests that certain models suffer from severe internal probability biases when processing clinical texts. MaKE’s Centering module effectively rectifies these biases, ensuring robust performance across diverse backbones without negatively affecting already-calibrated models.

6.2.3 Robustness Analysis on MHP-Eval-CF. To verify whether MaKE truly learns clinical concepts rather than relying on superficial shortcuts, we evaluated its performance on the **MHP-Eval-CF** counterfactual dataset. For this experiment, we compared MaKE against the direct inference baseline across six representative models. Notably, for the *Qwen series*, we selected the latest and most capable Qwen3-8B as the representative backbone to rigorously test the framework’s upper limits.

Table 7 presents the comparative results. MaKE demonstrates exceptional robustness, achieving consistent positive improvements across all models and tasks.

- **Screening & Severity:** The framework delivers transformative gains in these tasks. In Screening, MaKE nearly doubles the performance of baselines (e.g., boosting Gemma-3-1B by **+47.36%**), proving its resistance to noise and keyword adversarial attacks. Similarly, in Severity Assessment, models like Llama-3.1-8B and Vicuna-7B see massive improvements of over **50%**, indicating that our retrieval-augmented calibration successfully guides models to adhere to granular severity criteria even under perturbation.

Table 7. Robustness comparison on **MHP-Eval-CF**. We report Accuracy (%) for the Baseline (Zero-shot) and MaKE (Ours). Δ denotes the absolute performance gain. Positive improvements are marked in **green**.

Model	Screening			Disorder Identification			Severity Assessment		
	Baseline	Ours	Δ	Baseline	Ours	Δ	Baseline	Ours	Δ
Gemma-3-1B	44.92	92.28	+47.36	35.77	36.20	+0.43	27.05	27.49	+0.44
Llama-3.1-8B	52.21	95.03	+42.82	50.53	72.41	+21.88	26.85	78.20	+51.35
Mistral-7B-v0.3	43.49	94.07	+50.58	51.83	57.67	+5.84	27.23	49.76	+22.53
Phi-3-mini-4k	50.15	92.55	+42.40	52.18	77.05	+24.87	30.09	72.99	+42.90
Qwen3-8B	52.62	93.79	+41.17	50.89	53.64	+2.75	27.56	50.24	+22.68
Vicuna-7B-v1.5	47.90	92.97	+45.07	38.37	74.55	+36.18	20.54	73.46	+52.92

- **Disorder Classification:** MaKE continues to provide stable gains in the most challenging classification task. While strong baselines like Qwen3-8B [66] already possess high capability, MaKE further enhances its robustness ($\Delta = +2.75\%$), avoiding the regression often seen in adaptation methods. For models like Vicuna-7B, the improvement remains substantial (**+36.18%**). This universal enhancement confirms that MaKE effectively mitigates the "brittleness" of standard LLMs, ensuring reliable decision-making in high-stakes mental health sensing.

7 Conclusion

In conclusion, our study shows that robust mental health sensing with LLMs cannot be assessed by final answers alone. Through MHP-Eval, we demonstrate that meaning-preserving counterfactual edits can induce substantial and practically consequential shifts in model judgments, with vulnerability varying across task types. By separating internal knowledge from external knowledge, we reveal a consistent knowing-but-not-saying pattern, where models retain task-relevant evidence in their representations yet fail to express it reliably under perturbations. Building on this insight, MaKE provides a parameter-free inference-time strategy to leverage internal evidence for more stable decisions, yielding clear robustness gains without additional model training. Together, these results argue for robustness stress tests and internal evidence analysis as essential components for deploying LLM-based mental health sensing systems, and motivate future work on principled calibration, safety-aware interventions, and broader evaluations across settings and populations.

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A Appendix

A.1 Label-Space Confusion in Cross-Task Evaluation (Mental-LLM)

Although Mental-LLM [65] performs strongly on the tasks it was explicitly aligned with (Screening and Severity), we observe a systematic failure mode when transferring to other task schemas. As shown in Figure 9, when prompted for *Disorder Classification* with a fixed label set, the model outputs labels from an unrelated space (e.g., severity-/risk-style labels such as Attempt or severe). Similarly, for open-ended *Reasoning* prompts that require short free-form attribution analysis, it sometimes collapses back to classification-style outputs (e.g., Ideation or Yes). This suggests that standard domain instruction tuning may overfit to superficial input–output templates, leading to brittle cross-task generalization and instruction-unfaithful behavior.

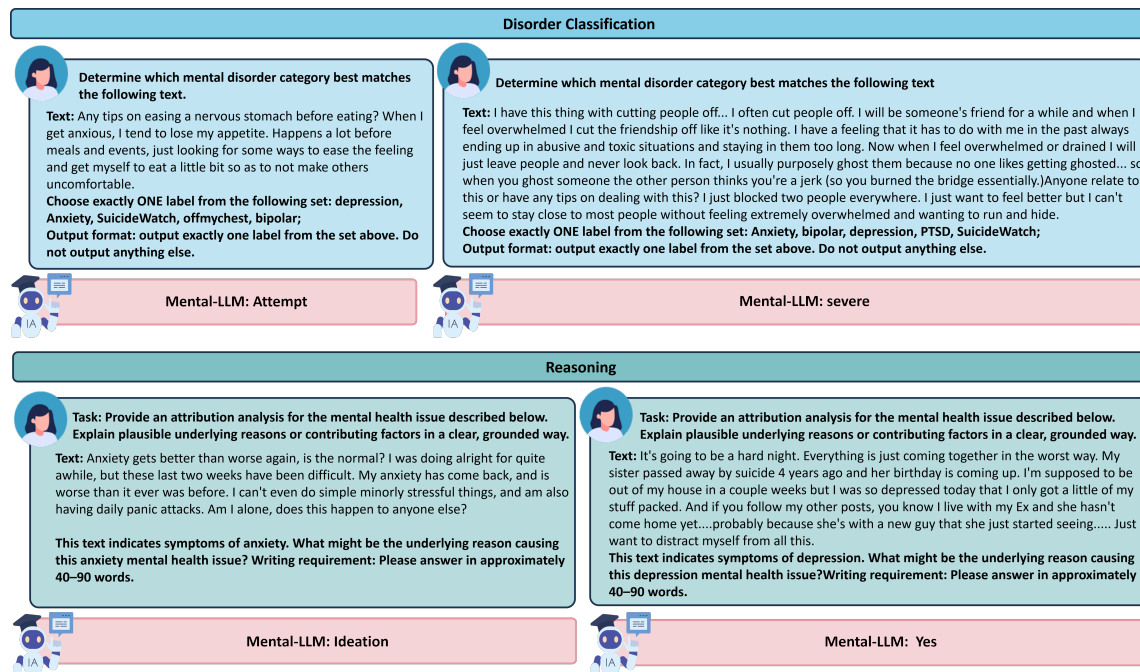


Fig. 9. Failure cases of Mental-LLM under cross-task prompts.

A.2 Full List of Prompt Templates

Table 8 details the complete set of 5 prompt templates designed for each task in MHP-Eval-Base. During the construction of the benchmark, for each data sample, one of these five templates is randomly selected to wrap the input text. This design ensures that the evaluation reflects the model's ability to understand the core clinical intent rather than overfitting to a specific phrasing.

Received 20 February 2007; revised 12 March 2009; accepted 5 June 2009

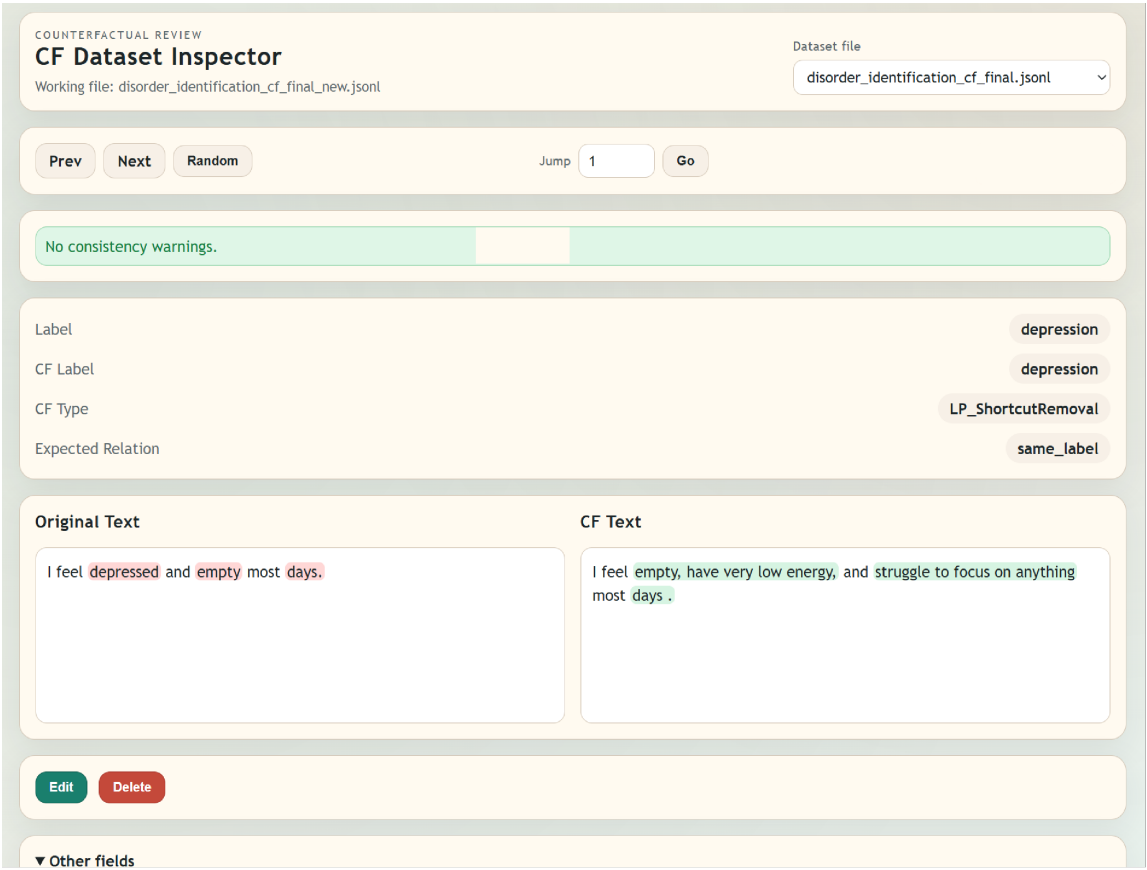


Fig. 10. MHP-Eval-CF Annotation tools.

Table 8. The complete set of 5 prompt templates used for each task in MHP-Eval-Base.

Task	Prompt Template Variations
Screening	1. "Is there an indication of psychological stress in the following text? Answer Yes or No." 2. "Analyze the user's post. Does the content suggest the presence of mental distress?" 3. "Read the text below and determine if the author is exhibiting signs of mental health issues. Output 'Yes' or 'No'." 4. "Classify the following post based on whether it indicates psychological stress. Please answer with Yes or No." 5. "Does the provided text contain evidence of mental distress or psychological strain? Answer Yes/No."
Disorder Cls.	1. "Which mental health category best fits the expressed symptoms? Choose one from: [List]." 2. "Diagnose the most likely condition described in the text based on the standard DSM-5 categories." 3. "Identify the specific mental health disorder characterized by the symptoms in the text. Select from the following options: [List]." 4. "Based on the clinical signs in the post, categorize the user's condition into one of these labels: [List]." 5. "Analyze the symptoms described. Which of the following mental health disorders is the most probable diagnosis? Options: [List]."
Severity Ass.	1. "What is the severity level of the expressed symptoms? (Minimum/Mild/Moderate/Severe)" 2. "Rate the intensity of the user's mental health condition on a four-point scale." 3. "Assess the severity of the mental health symptoms described in the text. Choose one: Minimum, Mild, Moderate, or Severe." 4. "How severe are the psychological symptoms presented in this post? Please classify as Minimum, Mild, Moderate, or Severe." 5. "Determine the intensity level of the distress shown. Output one of the following levels: Minimum, Mild, Moderate, Severe."
Reasoning	1. "What underlying factors may explain the expressed mental health symptoms?" 2. "Provide a clinical attribution analysis. Why does the user appear to be experiencing this condition?" 3. "Explain the reasoning behind the mental health status identified in the text. What specific details support this conclusion?" 4. "Analyze the text and describe the potential causes or contributing factors for the user's mental health state." 5. "Generate a brief clinical explanation clarifying why the text indicates specific mental health symptoms."